Preet Talati

talatipreet@gmail.com 8478771107 Linkedln Portfolio

PROFESSIONAL SUMMARY

I am a software engineer with expertise in building scalable full-stack systems, secure government platforms, and AI-powered developer tooling across C#, .NET, React, TypeScript, and Python. Proven track record taking products from 0 to 1, architecting cloud platforms serving 10,000+ users, and leading stakeholder-facing technical delivery.

CORE SKILLS

- **Applied AI & Automation** — LLMs, prompt engineering, RAG patterns, semantic search tuning, agent workflows, MCP-based AI tooling
- **Integrations & Workflows** — REST/GraphQL APIs, webhooks, event-driven/pub-sub systems, SaaS integrations, data synchronization
- **Frontend Engineering** — React, TypeScript, JavaScript, Redux, responsive UI development
- **Backend & Cloud** — .NET Core, ASP.NET Core, Node.js/Express, Python, serverless (Firebase Cloud Functions, AWS Lambda, GCP Functions), CI/CD pipelines
- **Data & Modeling** — SQL Server (T-SQL), MySQL, Firestore, schema design, data mapping, normalization, query optimization
- **Architecture & Systems** — microservices, modular domain design, RBAC frameworks, design patterns (MVC, Repository, Factory, DI)
- **Collaboration & Delivery** — cross-functional coordination, customer-facing engineering, live demos, rapid prototyping, technical enablement

EDUCATION

University of Illinois at Chicago, *Bachelor of Science in Computer Science*
- Honors: Summa Cum Laude, GPA: 3.92/4.0

PROFESSIONAL EXPERIENCE

Senior Software Engineer, *Defense Language Institute Foreign Language Center* Aug 2026 – Present
- **Led end-to-end project delivery** as the primary technical point of contact for government stakeholders, translating operational requirements into technical roadmaps, scoped deliverables, and production releases.
- **Drove solution architecture and technical decisions**, evaluating existing enterprise systems versus greenfield development based on security, cost, integration complexity, and long-term maintainability.
- **Architected full-stack C#/.NET applications and SQL Server data models**, improving query performance by **60%** through database design and optimization.
- **Owned application security architecture**, implementing enterprise identity integrations, custom **CAC/PKI** authentication, certificate validation, claims processing, and **RBAC** across **45+ roles**.
- **Coordinated secure production releases** across government stakeholders, **QA**, Security, and **DevOps**, overseeing **GitLab CI/CD** controls, vulnerability scanning, and IIS deployment workflows.

Software Engineer, *Defense Language Institute Foreign Language Center* Aug 2024 – Sep 2026
- **Developed full-stack applications** using C#, ASP.NET Core MVC, REST APIs, and SQL Server, delivering production functionality from requirements through deployment.
- **Built greenfield applications and enterprise integrations** across **Azure**, **Azure DevOps**, **Dynamics**/**Dataverse**, **Microsoft 365**, and **Active Directory** using **REST APIs** and relational data models.
- **Implemented authentication and authorization** using Windows **Authentication**, **Entra ID**, **Okta OIDC**, and custom **CAC**/**PKI workflows**, supporting **45+ RBAC roles**.
- **Automated CI/CD workflows** with **GitLab**, PowerShell runners, automated testing, **SAST**, and dependency scanning for **IIS** deployments across DEV, STAGING, and PROD.
- **Improved SQL Server query performance by ~25%** through indexing and **T-SQL** optimization; developed a Python BFS crawler with authenticated access, multilingual keyword matching, and automated Excel reporting.
- **Applied MVC, Repository, Factory, and Dependency Injection patterns** to improve separation of concerns, testability, and maintainability, **documenting** architecture and integration decisions.

Founding Engineer , *Gyfr* Jun 2024 – Feb 2025
- **Owned** technical vision and product delivery for the web and mobile platform, from MVP to production.
- **Architected** and shipped the backend **API** on Firebase Cloud **Functions (Node.js, TypeScript, Express)**. It runs sessions, subscriptions, and payments, and the platform grew to **10,000+ users**.
- **Built** event-driven billing with **Stripe webhooks**, including **idempotency** handling, subscription lifecycle management and billing-state sync. I also built deferred plan changes aligned to billing cycles.
- **Centralized** the **React/Redux/Axios API** client layer across **50**+ routes and split the architecture into client, trainer, session and subscription domains, which made new features faster to build.
- **Added** metrics **instrumentation** and used it to tune the app, improving performance by **40%.**

PROJECTS

No2SQL | C# | .NET Core | AI Developer Tooling
- **Built** a C# **MCP server and NuGet package** using **AI-assisted development** to automate **MongoDB-to-SQL schema** discovery, relationship inference, and migration generation.
- **Designed LLM application workflows and an MCP server** with prompt-driven schema discovery and **tool orchestration**, exposing **MCP capabilities to GitHub Copilot, VS Code, and Claude Desktop.**
- **Developed** a **relationship inference engine** that analyzes dynamic **JSON** documents to discover **cross-collection relationships**, **normalize** data models, and generate deterministic **SQL** schemas.
- **Developed** a deterministic **SQL** generation pipeline that creates **migration**-ready **DDL**, **SQL seed data**, foreign key constraints, and ER diagrams.
- **Developed tooling** to **automate** data **migration** between **NoSQL** and **relational databases.**